

BHARATH GENJI MOHANARANGA

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Summary

AI Engineer who ships production multi-agent systems that move real business metrics — cut 15-minute workflows to under 2 minutes, recovered 10K+ analyst hours, and drove \$2M+ in projected savings at NRG Energy. 4+ years spanning the full stack from LangGraph/ADK agent orchestration and RAG pipelines to cross-cloud deployments on GCP, AWS, and Azure with end-to-end MLOps. Equally comfortable designing LLM architectures (ReAct agents, structured tool calling, guardrails) and wiring them into enterprise backends (SAP, Microsoft Graph, Vertex AI). Thrive in zero-to-one environments — first engineer at a document-intelligence startup, lead developer on two greenfield AI platforms.

Technical Skills

Languages & Frameworks: Python; FastAPI; Flask-RESTX; OpenCV; PySpark

GenAI & LLMs: LangGraph; LangChain (ReAct agents); Google ADK; RAG; Weaviate; pgvector; AlloyDB; OpenAI; Gemini 2.5 Flash; Google Document AI; Azure Document Intelligence; Stable Diffusion

Cloud & Infra: GCP (Vertex AI, Cloud Run, Firestore, BigQuery, Secret Manager); AWS (ECS Fargate, Lambda, SQS, EventBridge); Azure (Container Apps, SQL, Key Vault, Blob Storage, DevOps); Microsoft Graph API (SharePoint, Email); Cloudflare Zero Trust

Data & Streaming: Snowflake; Teradata; HBase; Hive; Kafka; YARN; Hadoop; MapReduce; Tableau

MLOps & DevOps: Docker; Terraform; GitHub Actions; Azure DevOps CI/CD; secrets management; observability (traces/latency/cost); Bitbucket

Relevant Work Experience

AI Consultant — Data & AI Team

NRG Energy

Sep 2025 – Present

Houston, TX (Hybrid)

- Lead developer of EMMA BPEM (Enhanced Message Management Analysis), an AI-powered customer service automation system using LangGraph ReAct agents with Gemini 2.5 Flash, reducing case resolution time from 15–30 min to 1–3 min across 1,000+ monthly cases.
- Architected multi-agent orchestration with specialized nodes (Entry/Shielding, Orchestrator, Supervisor, Specialist agents) handling 5 use cases: referral credits, digital/physical gift cards, wrong amounts, and Amazon incentives — achieving 95% accuracy in pilot (1,500 cases).
- Built structured tool-calling integrations with SAP SOAP/OData APIs, TSC Portal, Blackhawk, and Datalake for autonomous case investigation and resolution with human-in-the-loop checkpoints.
- Deployed production services on Google Cloud Run with Firestore checkpointing, Google Secret Manager, and Azure DevOps CI/CD — projecting \$2M annual savings and 10,000+ analyst hours recovered.
- Developing Credit Risk Review Agent using Google ADK with multi-agent architecture (Credit Risk Agent + Research Agent) to automate Credit Review Templates — handling financial spreading, ratio calculations, debt maturity analysis, covenant extraction, and early-warning detection.
- Engineered cross-cloud authentication between Azure Container Apps and Google Vertex AI using service account keys stored in Azure Key Vault, enabling seamless multi-cloud agent deployment.
- Built Microsoft Graph API integrations for SharePoint document management and automated email notifications via client credentials flow (MSAL), integrated as MCP tools in the NRG Tool Factory platform.
- Implemented RAG pipelines with AlloyDB vector store (text-embedding-004) via Knowledge Enablement services for credit policy retrieval and financial document grounding.

ST Hodge Fellow / AI & Data Science Intern

Kingsman Academy Public Charter School

Feb 2025 – Dec 2025

Washington, DC (Hybrid)

- Architected a unified data & GenAI platform ingesting LMS, finance, and attendance into a central feature store and vector store, powering real-time dashboards and LLM tools.
- Built LLM copilots (RAG over policy/IEPs/attendance) with LangChain ReAct/tool use; assistants draft interventions, summarize histories, and return cited answers.
- Productionized prompts & evals: prompt/version control, dataset-grounded evaluation, guardrails (PII redaction, refusal rules), and observability (traces, latency, token/cost).
- Deployed low-latency Python APIs on AWS (ECS/Lambda) and GCP (Vertex AI + Cloud Run) with batching/caching to reduce tail latency and compute spend.
- Implemented semantic search via Weaviate/pgvector for evidence-grounded retrieval; delivered Tableau dashboards with an “Explain this trend” LLM action.

Founding AI Software Engineer

TheDetail.ai

May 2025 – Jul 2025

Washington, DC (Remote)

- Spearheaded an enterprise GenAI document-intelligence platform as first engineer; designed FastAPI microservices on AWS ECS Fargate with Lambda/SQS/EventBridge; scaled to 10K+ docs/min (p99 < 200 ms).
- Built ensemble extraction using Google Document AI, Azure Document Intelligence, and Gemini LLMs; achieved F1 0.97 across 40+ drawing formats.
- Implemented Weaviate + OpenAI vector layer with semantic re-ranking (<50 ms retrieval over 30M vectors); orchestrated RAG workflows with LangChain ReAct agents.
- Established end-to-end MLOps (GitHub Actions, Terraform, Docker); deployed GPU inference bursts on GCP Vertex AI, reducing unit cost by 18%.

AI/ML Engineer Intern

Infotrend

Jun 2024 – Aug 2024

Washington, DC

- Deployed real-time image-processing stack using Dockge; integrated Stable Diffusion with Open WebUI; secured access via VPS + Cloudflare Zero Trust.
- Engineered CV/NLP apps with Python, TensorFlow, PyTorch, OpenCV; summarized state-of-the-art deep & RL research to guide model and infra choices.

Research Assistant

George Washington Institute of Public Policy

May 2024 – Aug 2024

Washington, DC

- Built analytics app with LAiSER library to reorganize skill data from NLx; automated job-data analysis across Texas industries, increasing handling efficiency by 40%.

Executive Cloud & Data Engineer

InfoCepts

Apr 2023 – Jul 2023

Chennai, India

- Integrated Flask-RESTX APIs with Azure SQL and AI analytics, boosting productivity by 50%; standardized Bitbucket workflows and led Agile delivery.

Associate Cloud & Data Engineer

InfoCepts

Sep 2021 – Apr 2023

Chennai, India

- Modernized Order-Management System on AWS, cutting manual effort by 25%; built automated Spark quality analyzer improving assessment performance by 30%.
- Implemented automated ETL for Teradata → Snowflake migration, reducing manual effort by 50%.

Intern, Cloud & Data Engineer

InfoCepts

Mar 2021 – Aug 2021

Chennai, India

- Engineered high-performance ETL tool using HBase, Hive, Kafka, YARN, Hadoop, MapReduce, PySpark; reduced processing time by 40%.

Technical Projects

Enhancing Park Safety — Data-Driven Analysis in NYC Parks

Jan 2024 – Mar 2024

- Analyzed NYC park-crime data (2015–2023) with Python and Tableau; delivered safety recommendations based on park size and incident trends.

NCAA Tournament Prediction — GWU AI/ML Bracket Challenge

Jan 2024 – Mar 2024

- Trained XGBoost model on 21 years of NCAA data (Brier score 0.1759); ranked top 2.2% nationally.

Accomplishments

- First Runner-up at InfoCepts E360 booth for pioneering content (product ideation and development).
- Winner, ITC & GWU AI/ML Bracket Challenge (overall score 1370) through innovative ML methodologies.

Education

M.S. in Data Science

The George Washington University

Aug 2023 – May 2025

GPA: 3.8

B.Tech. in Computer Science

SRM Institute of Science and Technology

Jul 2017 – Jun 2021

GPA: 3.4